



**NATIONAL
SENIOR CERTIFICATE/
NASIONALE
SENIOR SERTIFIKAAT**

GRADE/GRAAD 11

NOVEMBER 2022

**MATHEMATICS P1/WISKUNDE V1
MARKING GUIDELINE/NASIENRIGLYN**

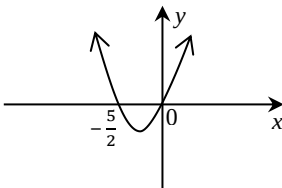
MARKS/PUNTE: 150

This marking guideline consists of 18 pages./
Hierdie nasienriglyn bestaan uit 18 bladsye.

NOTE/LET WEL:

- If a candidate answers a question TWICE, mark the FIRST attempt ONLY.
Indien 'n kandidaat 'n vraag TWEE keer beantwoord, merk SLEGS die EERSTE poging.
- Consistent accuracy applies in ALL aspects of the marking guideline.
Volgehoue akkuraatheid geld deurgaans in ALLE aspekte van die nasienriglyn.
- If a candidate crossed out an attempt of a question and did not redo the question, mark the crossed-out attempt.
Indien 'n kandidaat 'n poging vir 'n vraag deurgetrek het en nie die vraag weer beantwoord het nie, merk die poging wat deurgetrek is.
- The mark for substitution is awarded for substitution into the correct formula.
Die punt vir substitusie word toegeken vir substitusie in die korrekte formule.

QUESTION 1/VRAAG 1

1.1.1	$x^2 + 5x - 6 = 0$ $(x+6)(x-1) = 0$ $\therefore x = -6 \text{ or / of } x = 1$	✓ factors / faktore ✓ ✓ answers / antwoorde (3)
1.1.2	$5x^2 + x - 3 = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $= \frac{-1 \pm \sqrt{1^2 - 4(5)(-3)}}{2(5)}$ $= \frac{-1 \pm \sqrt{61}}{10}$ $= 0,68 \text{ or / of } -0,88$	✓ substitution / vervanging ✓ ✓ answers / antwoorde (3)
1.1.3	$(2x-1)(x+3) \geq -3$ $2x^2 + 5x - 3 \geq -3$ $2x^2 + 5x \geq 0$ $x(2x+5) \geq 0$ $\therefore x \leq -\frac{5}{2} \text{ or / of } x \geq 0$	 ✓ $2x^2 + 5x - 3$ ✓ standard form / standaardvorm ✓ factors / faktore ✓ ✓ answers / antwoorde (5)

1.1.4	$\sqrt{x} - \sqrt{x-5} = 1$ $\sqrt{x} - 1 = \sqrt{x-5}$ $(\sqrt{x} - 1)^2 = (\sqrt{x-5})^2$ $x - 2\sqrt{x} + 1 = x - 5$ $6 = 2\sqrt{x}$ $(6)^2 = (2\sqrt{x})^2$ $\therefore 4x = 36$ $x = 9$	$\checkmark \sqrt{x} - 1 = \sqrt{x-5}$ $\checkmark \text{ squaring both sides}$ $\text{kwadreer beide kante}$ $\checkmark x - 5 = x - 2\sqrt{x} + 1$ $\checkmark \text{ answer / antwoord}$
1.2	$2x - y = 1 \quad (1)$ $y^2 - xy = x + 7 \quad (2)$ $y = 2x - 1 \quad (3)$ Subst.(3) into 2 / Verv.(3) in (2) $(2x - 1)^2 - x(2x - 1) = x + 7$ $4x^2 - 4x + 1 - 2x^2 + x - x - 7 = 0$ $2x^2 - 4x - 6 = 0$ $x^2 - 2x - 3 = 0$ $(x - 3)(x + 1) = 0$ $\therefore x = 3 \text{ or / of } x = -1$ $y = 2(3) - 1 \text{ or / of } y = 2(-1) - 1$ $\therefore y = 5 \text{ or / of } y = -3$ OR/OF $2x - y = 1 \quad (1)$ $y^2 - xy = x + 7 \quad (2)$ $x = \frac{1+y}{2} \quad (3)$ Subst.(3) into 2 / Verv.(3) in (2) $y^2 - y\left(\frac{1+y}{2}\right) = \left(\frac{1+y}{2}\right) + 7$ $2y^2 - y(1+y) = (1+y) + 14$ $2y^2 - y - y^2 = 1 + y + 14$ $y^2 - 2y - 15 = 0$ $(y - 5)(y + 3) = 0$ $\therefore y = 5 \text{ or / of } y = -3$ $x = \frac{1+5}{2} \text{ or / of } x = \frac{1-3}{2}$ $\therefore x = 3 \text{ or / of } x = -1$	$\checkmark y = 2x - 1$ $\checkmark \text{ substitution / vervanging}$ $\checkmark \text{ standard form / standaardvorm}$ $\checkmark \text{ factors / faktore}$ $\checkmark \text{ both x-values / beide x-waardes}$ $\checkmark \text{ both y-values / beide y-waardes}$ $\checkmark x = \frac{1+y}{2}$ $\checkmark \text{ substitution / vervanging}$ $\checkmark \text{ standard form / standaardvorm}$ $\checkmark \text{ factors / faktore}$ $\checkmark \text{ both y-values / beide y-waardes}$ $\checkmark \text{ both x-values / beide x-waardes}$

(4)

(6)

1.3	$\hat{C} = 90^\circ$ (angle in a semi-circle) (hoek in halwe sirkel) $\therefore$ By Pythagoras's Theorem Stelling van Pythagoras : $AB^2 = AC^2 + BC^2$ $= (x+3)^2 + (5-x)^2$ $= x^2 + 6x + 9 + 25 - 10x + x^2$ $= 2x^2 - 4x + 34$ $= 2(x^2 - 2x + 17)$ $= 2(x^2 - 2x + 1 - 1 + 17)$ $= 2(x-1)^2 + 32$ $\therefore x = 1$ OR/OF AB is minimum when AB^2 is minimum $AB^2 = (x+3)^2 + (5-x)^2$ $= x^2 + 6x + 9 + 25 - 10x + x^2$ $= 2x^2 - 4x + 34$ AB^2 is minimum at / AB^2 is 'n minimum by: $x = -\frac{b}{2a}$ $= \frac{-(-4)}{2(2)}$ $= 1$	✓ $\hat{C} = 90^\circ$ ✓ substitution / vervanging ✓ $2x^2 - 4x + 34$ ✓ completing the square vierkantsvoltooiing ✓ answer / antwoord ✓ statement / stelling ✓ substitution / vervanging ✓ $2x^2 - 4x + 34$ ✓ $x = -\frac{b}{2a}$ ✓ answer / antwoord (5)
		[26]

QUESTION 2/VRAAG 2

2.1	$\frac{2^{2x} - 4^{x+1}}{4^x + 2^{2x-1}}$ $= \frac{2^{2x} - 2^{2x+2}}{2^{2x} + 2^{2x-1}}$ $= \frac{2^{2x} - 2^{2x} \cdot 2^2}{2^{2x} + 2^{2x} \cdot 2^{-1}}$ $= \frac{2^{2x}(1 - 2^2)}{2^{2x}(1 + 2^{-1})} \text{ or / of } \frac{2^{2x}(1 - 4)}{2^{2x}(1 + \frac{1}{2})}$ $= \frac{-3}{\frac{3}{2}}$ $= -2$	✓ 2^{2x+2} and/en 2^{2x} ✓ inverse of exp. law <i>omgekeerde van eksp. wet</i> ✓ factorisation of numerator <i>faktorisering van teller</i> ✓ answer / antwoord
		(4)
2.2.1	$3x^{\frac{3}{2}} = 81$ $x^{\frac{3}{2}} = 27$ $\left(x^{\frac{3}{2}}\right)^{\frac{2}{3}} = (27)^{\frac{2}{3}}$ $\therefore x = 9$	✓ $x^{\frac{3}{2}} = 27$ ✓ $\left(x^{\frac{3}{2}}\right)^{\frac{2}{3}} = (27)^{\frac{2}{3}}$ ✓ answer / antwoord
		(3)
2.2.2	$2^x + 5 = 3 \cdot 2^{1-x}$ $2^x + 5 = 3 \cdot 2 \cdot 2^{-x}$ $2^x + 5 = \frac{3 \cdot 2}{2^x}$ $(2^x)^2 + 5 \cdot 2^x - 6 = 0$ $(2^x + 6)(2^x - 1) = 0$ $\therefore 2^x \neq -6 \quad \text{or / of} \quad 2^x = 1$ $2^x = 2^0$ $\therefore x = 0$	✓ inverse of exp. law <i>omgekeerde van eksp. wet</i> ✓ multiplying by 2^x <i>maal met 2^x</i> ✓ factors / faktore ✓ both answers/beide antwoorde ✓ selection / keuse
		(5)

2.3	$\frac{1+\sqrt{2}}{3+2\sqrt{2}}$ $= \frac{(1+\sqrt{2})(3-2\sqrt{2})}{(3+2\sqrt{2})(3-2\sqrt{2})}$ $= \frac{3-2\sqrt{2}+3\sqrt{2}-2.2}{3^2-(2\sqrt{2})^2}$ $= \frac{\sqrt{2}-1}{9-8}$ $= \sqrt{2}-1$ $\therefore a=2, \quad b=-1$	✓ rationalising the denominator <i>rasionalisering van die noemer</i> ✓ simplification / <i>vereenvoudiging</i> ✓ $\sqrt{2}-1$ ✓ $a=2$ ✓ $b=-1$ (5)
		[17]

QUESTION 3/VRAAG 3

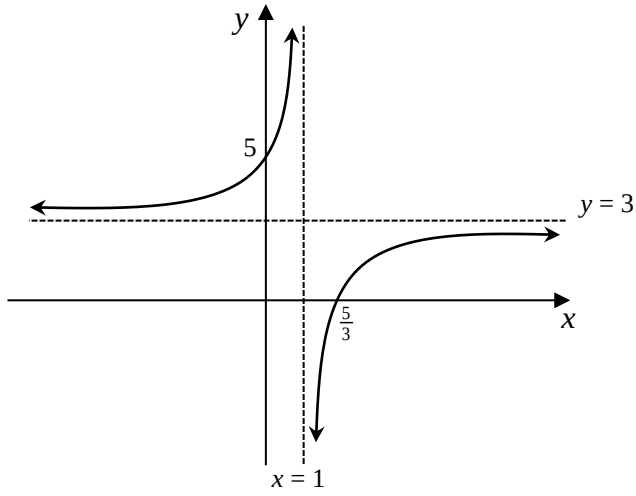
3.1.1	$-2 ; 3 ; 8 ; \dots$ $T_n = 5n - 7$	$\checkmark 5n \quad \checkmark -7$ (2)
3.1.2	$T_n = 5n - 7$ $T_{18} = 5(18) - 7$ $= 83$	$\checkmark$ substitution / <i>vervang</i> $\checkmark 83$ (2)
3.1.3	$T_n = 5n - 7$ $473 = 5n - 7$ $480 = 5n$ $\therefore n = 96$	$\checkmark$ substitution / <i>vervang</i> $\checkmark$ answer / <i>antwoord</i> (2)

3.2	$T_{11} = -19$ $T_{23} = 65$ $-19 ; a ; b ; c ; e ; \dots ; 65$ $a - (-19) = b - a = c - b = \dots = d$ $\therefore$ There are 12 common differences / <i>Daar is 12 gemeenskaplike verskille</i> $\therefore 12d = 65 - (-19)$ $12d = 84$ $\therefore d = 7$ $\therefore T_n = 7n + b$ $-19 = 7(11) + b$ or / of $65 = 7(23) + b$ $\therefore b = -96$ $\therefore T_n = 7n - 96$ For negative terms : $T_n < 0$ <i>Vir negatiewe terme :</i> $\therefore 7n - 96 < 0$ $7n < 96$ $\therefore n < 13,71$ $\therefore$ Number of negative terms = 13 <i>Aantal negatiewe terme</i> OR/OF $12d = 65 - (-19)$ $d = \frac{84}{12}$ $= 7$ But / <i>Maar</i> : T_1 to T_{11} are all negative / <i>almal negatief</i> $T_{12} = -19 + 7 = -12$ $T_{13} = -12 + 7 = -5$ $T_{14} = -5 + 7 = 2$ $\therefore$ There are 13 negative terms <i>Daar is 13 negatiewe terme</i>	$\checkmark 12d = 65 - (-19)$ $\checkmark d = 7$ $\checkmark T_n = 7n - 96$ $\checkmark 7n - 96 < 0$ $\checkmark n = 13$ OR/OF $\checkmark 12d = 65 - (-19)$ $\checkmark d = 7$ $\checkmark T_1$ to T_{11} are all negative / <i>is almal negatief</i> $\checkmark T_{12} = -12 \ \& \ T_{13} = -5$ $\checkmark$ answer / <i>antwoord</i> (5)
		[11]

QUESTION 4/VRAAG 4

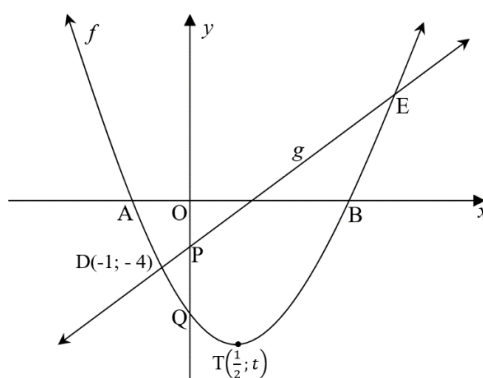
4.1	$ \begin{array}{ccccccc} 204 & ; & 176 & ; & 150 & ; & 126 \\ & \swarrow & \nearrow & \swarrow & \nearrow & \swarrow & \nearrow \\ & -28 & & -26 & & -24 & \\ & \searrow & \swarrow & \searrow & \swarrow & \searrow & \swarrow \\ & & 2 & & 2 & & \end{array} $ 104 ; 84	✓ 104 ✓ 84 (2)
4.2	$ \begin{array}{lll} 2a = 2 & 3a + b = -28 & a + b + c = -12 \\ \therefore a = 1 & 3(1) + b = -28 & 1 - 31 + c = 204 \\ & b = -31 & c = 234 \\ \\ \therefore T_n = n^2 - 31n + 234 \end{array} $	✓ $a = 1$ ✓ $b = -31$ ✓ $c = 234$ ✓ $T_n = n^2 - 31n + 234$ (4)
4.3	$ \begin{array}{l} n^2 - 31n + 234 = 36 \\ n^2 - 31n + 198 = 0 \\ n = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \text{or / of } (n-9)(n-22) = 0 \\ = \frac{-(-31) \pm \sqrt{(-31)^2 - 4(1)(198)}}{2(1)} \\ = \frac{31 \pm \sqrt{169}}{2} \\ \therefore n = 9 \quad \text{or / of } n = 22 \end{array} $	✓ $n^2 - 31n + 234 = 36$ ✓ subst. into formule / factors verv. in formule / faktore ✓ $n = 9$ ✓ $n = 22$ (4)
4.4	$ \begin{array}{l} n^2 - 31n + 234 = 0 \quad \text{or / of } (n-13)(n-18) = 0 \\ n = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ = \frac{-(-31) \pm \sqrt{(-31)^2 - 4(1)(234)}}{2(1)} \\ = \frac{31 \pm \sqrt{25}}{2} \\ \therefore n = 13 \quad \text{or / of } n = 18 \\ \\ \therefore T_{14} \text{ \& } T_{17} = -4 \\ T_{15} \text{ \& } T_{16} = -6 \end{array} $	✓ $T_n = 0$ ✓ method / metode subst. into formule / factors verv. in formule / faktore ✓ $n = 13$ and / en $n = 18$ ✓ -4 ✓ -6 (5)
		[15]

QUESTION 5/VRAAG 5

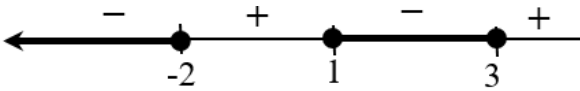
5.1	$x = 1$ $y = 3$	$\checkmark x = 1$ $\checkmark y = 3$ (2)
5.2	$0 = \frac{-2}{x-1} + 3$ $\frac{2}{x-1} = 3$ $3(x-1) = 2$ $x = \frac{5}{3}$ $y = \frac{-2}{x-1} + 3$ $= \frac{-2}{0-1} + 3$ $= 5$ $\therefore$ Intercepts are at / <i>Afsnitte is by:</i> $(\frac{5}{3}; 0)$ and / <i>en</i> $(0; 5)$	$\checkmark y = 0$ $\checkmark x = \frac{5}{3}$ $\checkmark y = 5$ (3)
5.3		$\checkmark$ x-intercept / <i>x-afsnit</i> $\checkmark$ y-intercept / <i>y-afsnit</i> $\checkmark$ asymptotes / <i>asimptote</i> $\checkmark$ shape & quadrants <i>vorm & kwadrante</i> (4)
5.4	$y = -(x-1) + 3$ $= -x + 4$	$\checkmark\checkmark y = -x + 4$ (2)

5.5	$g(x) = -x + b$ $-2 = -(5) + b$ OR/OF $y - y_1 = m(x - x_1)$ $\therefore b = 3$ $y + 2 = -1(x - 5)$ $g(x) = -x + 3$ $\therefore y = g(x) = -x + 3$	$\checkmark a = -1$ $\checkmark$ substitution / <i>vervanging</i> $\checkmark b = 3$ (3)
5.6	$f(x) = g(x)$ $\frac{-2}{x-1} + 3 = -x + 3$ $\frac{-2}{x-1} = -x$ $-x(x-1) = -2$ $-x^2 + x + 2 = 0$ $x^2 - x - 2 = 0$ $(x+1)(x-2) = 0$ $\therefore x = -1$ or / of $x = 2$ $\therefore y = -(-1) + 3$ or / of $y = -(2) + 3$ $= 4$ $= 1$ Points of intersection / <i>Snypunte by</i> : $(-1; 4)$ and / en $(2; 1)$ $\therefore d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $= \sqrt{(2 - (-1))^2 + (1 - 4)^2}$ $= \sqrt{18} = 3\sqrt{2}$	$\checkmark$ equating/ <i>gelykstel</i> : $f(x) = g(x)$ $\checkmark$ standard form / <i>standaardvorm</i> $\checkmark$ both sets of coordinates <i>beide pare van koördinate</i> $\checkmark$ substitution into correct formula / <i>vervanging in die</i> <i>korrekte formule</i> $\checkmark$ answer / <i>antwoord</i> (5)
5.7	$h(x) = -f(x+3)$ $= \frac{2}{(x+3)-1} - 3$ $= \frac{2}{x+2} - 3$	$\checkmark a = +2$ & $q = -3$ (reflection/ <i>refleksie</i>) $\checkmark x + 2$ (2)
		[21]

QUESTION 6/VRAAG 6



6.1	$f(x) = ax^2 + bx + c$ P is at/by: $(0; -2)$ Q is at/by: $(0; -6)$ $\therefore f(x) = ax^2 + bx - 6$ $-4 = (-1)^2 a + (-1)b - 6$ $2 = a - b \dots \dots \dots (1)$ At TP/By DP: $x = \frac{-b}{2a}$ $\frac{1}{2} = \frac{-b}{2a}$ $\therefore 2a = -2b$ $a = -b \dots \dots \dots (2)$ $\therefore 2 = -b - b$ $2 = -2b$ $b = -1$ $\therefore a = -(-1)$ $= 1$	$\checkmark c = -6$ $\checkmark a - b = 2$ $\checkmark$ substitute for x / vervang vir x $\checkmark a = -b$ $\checkmark$ values of a and b. waardes van a en b (5)
6.2	$f(x) = x^2 - x - 6$ $= x^2 - x + \frac{1}{4} - \frac{1}{4} - 6$ $= \left(x - \frac{1}{2}\right)^2 - 6\frac{1}{4}$ $t = -6\frac{1}{4}$ OR/OF $f(x) = x^2 - x - 6$ $\therefore y = \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right) - 6$ $= -6\frac{1}{4}$ $\therefore t = -6\frac{1}{4}$	$\checkmark$ completing the square vierkantsvoltooiing $\checkmark$ factorisation / faktorisering $\checkmark t = -6\frac{1}{4}$ OR/OF $\checkmark f\left(\frac{1}{2}\right)$ $\checkmark y = -6\frac{1}{4}$ $\checkmark t = -6\frac{1}{4}$ (3)

6.3	$f(x) = x^2 - x - 6$ $0 = (x-3)(x+2)$ $\therefore x = 3$ or / of $x = -2$ $\therefore A(-2;0)$ and/en $B(3;0)$	✓ factors / <i>faktore</i> ✓ $A(-2;0)$ ✓ $B(3;0)$ (3)
6.4	$f(x) = g(x)$ $x^2 - x - 6 = 2x - 2$ $x^2 - 3x - 4 = 0$ $(x-4)(x+1) = 0$ $\therefore x = 4$ or / of $x = -1$ $y = 2(4) - 2$ $= 6$ $\therefore E(4;6)$	✓ equating $f(x)$ and $g(x)$ <i>gelykstel van $f(x)$ en $g(x)$</i> ✓ standard form / <i>standaardvorm</i> ✓ x -values / x -waardes ✓ coordinates of E <i>koördinate van E</i> (4)
6.5	$y \geq -6\frac{1}{4}$ or / of $y \geq t$ $y \in [-6\frac{1}{4}; \infty)$ or / of $y \in [t; \infty)$	✓ ✓ answer / <i>antwoord</i> (2)
6.6	$g(x) = 2x - 2$ $\therefore 0 = 2x - 2$ $\therefore x = 1$ $x \leq -2$ or / of $1 \leq x \leq 3$ OR/OF  $\therefore x \leq -2$ or / of $1 \leq x \leq 3$	✓ $x \leq -2$ ✓ $1 \leq x \leq 3$ ✓ $x \leq -2$ ✓ $1 \leq x \leq 3$ (2)
		[19]

QUESTION 7/VRAAG 7

7.1	$f(x) = a^x + 1$ $9 = a^{-3} + 1$ $8 = a^{-3}$ $\therefore a^3 = \frac{1}{8}$ $\sqrt[3]{a^3} = \sqrt[3]{\frac{1}{8}}$ $\therefore a = \frac{1}{2}$	✓ substitution / <i>vervanging</i> ✓ $a^3 = \frac{1}{8}$ ✓ answer / <i>antwoord</i> (3)
7.2	$g(x) = -\left(\frac{1}{2}\right)^x + 1$	✓ $-\left(\frac{1}{2}\right)^x$ ✓ +1 (2)
		[5]

QUESTION 8/VRAAG 8

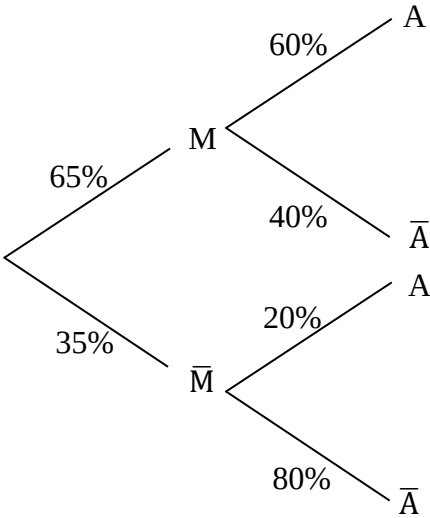
8.1	$1 + i_{eff} = \left(1 + \frac{i_{nom}}{12}\right)^{12}$ $1 + 0,0992 = \left(1 + \frac{x}{12}\right)^{12}$ $\therefore \sqrt[12]{1,0992} = \sqrt[12]{\left(1 + \frac{x}{12}\right)^{12}}$ $\therefore x = \left(\sqrt[12]{1,0992} - 1\right) \times 12$ $= 0,0950$ $\therefore$ The rate is 9,5% p.a. compounded monthly. <i>Die koers is 9,5% p.j. maandeliks saamgestel.</i>	✓ substitution / <i>vervanging</i> ✓ simplification / <i>vereenvoudiging</i> ✓ answer / <i>antwoord</i> (3)
8.2	$A = P(1 - i)^n$ $28607,30 = P(1 - 12\%)^7$ $P = \frac{A}{(1 - i)^n}$ $= \frac{28607,30}{(1 - 12\%)}$ $= R70\,000,00$	✓ correct formula <i>korrekte formule</i> ✓ substitution / <i>vervanging</i> ✓ answer / <i>antwoord</i> (3)

8.3.1	$A = P(1+i)^n$ $A = \left(\left(32\,000 \left(1 + \frac{0,086}{12} \right)^{36} \right) + 23\,000 \right) \left(1 + \frac{0,086}{12} \right)^{12}$ $= R70141,04$ OR/OF $A = P(1+i)^n$ $= 32\,000 \left(1 + \frac{0,086}{12} \right)^{48} + 23\,000 \left(1 + \frac{0,086}{12} \right)^{12}$ $= R70141,04$	$\checkmark i = \frac{0,082}{12}$ $\checkmark \left(32\,000 \left(1 + \frac{0,086}{12} \right)^{36} \right)$ $\checkmark +23\,000$ $\checkmark \times \left(1 + \frac{0,086}{12} \right)^{12}$ $\checkmark \text{ answer / antwoord}$ $\checkmark i = \frac{0,082}{12}$ $\checkmark \left(32\,000 \left(1 + \frac{0,086}{12} \right)^{48} \right)$ $\checkmark 23\,000 \left(1 + \frac{0,086}{12} \right)^{12}$ $\checkmark \text{ adding / optelling}$ $\checkmark \text{ answer / antwoord}$ (5)
8.3.2	$A = P(1+i)^n$ $= 70141,04 \left(1 + \frac{0,105}{4} \right)^8$ $= R86\,297,36$ $\therefore$ Loan / Lening : $R220\,000 - R86\,297,36$ $= R133\,702,64$	$\checkmark P = R70\,141,04$ $\checkmark \text{ substitution / vervanging}$ $\checkmark R86\,297,36$ $\checkmark \text{ answer / antwoord}$ (4)
		[15]

QUESTION 9/VRAAG 9

9.1.1	$P(A \text{ and/en } B) = 0$ $\therefore P(A \text{ or / of } B) = P(A) + P(B)$ $0,75 = 0,35 + P(B)$ $\therefore P(B) = 0,75 - 0,35$ $= 0,4 \text{ or / of } \frac{2}{5}$	✓ correct formula / korrekte formule ✓ substitution / vervanging ✓ answer / antwoord
9.1.2	$P(A \text{ and / en } B) = P(A) \times P(B)$ $\therefore P(A \text{ or/of } B) = P(A) + P(B) - P(A).P(B)$ $0,75 = 0,35 + P(B) - 0,35P(B)$ $0,4 = 0,65P(B)$ $\therefore P(B) = \frac{0,4}{0,65}$ $= \frac{8}{13}$	✓ correct formula / korrekte formule ✓ substitution / vervanging ✓ simplification / vereenvoudiging ✓ answer / antwoord
9.2.1		✓ 18, 12 and/en x (intersections) ✓ $34 - x$ ✓ x and/en y ✓ 8 and/en 11
9.2.2	$y + 12 + 11 + 18 = 81$ $y = 40$ $x + x + 11 + 12 + y + 18 + 34 - x + 8 = 130$ $x + 83 + 40 = 130$ $\therefore x = 7$	✓ y-value / y-waarde ✓ equation / vergelyking ✓ x-value / x-waarde
9.2.3	$P(\text{only one/slegs een}) = \frac{34-x}{130} + \frac{x}{130} + \frac{y}{130}$ $= \frac{27}{130} + \frac{7}{130} + \frac{40}{130}$ $= \frac{74}{130} = \frac{37}{65} \approx 0,57$	✓ method / metode ✓ answer / antwoord
		[16]

QUESTION 10/VRAAG 10

10.	 $P(A) = P(MA) + P(\bar{M}A)$ $= (65\% \times 60\%) + (35\% \times 20\%)$ $= 46\%$ $= \frac{23}{50}$	✓ $P(A) = P(MA) + P(\bar{M}A)$ ✓ substitution / vervanging ✓ answer / antwoord (5)
		[5]

TOTAL / TOTAAL: 150